Assignment 9

CS 202 Lab Assignment

Total Marks =20

Q1. Implement a simple counter mechanism that increments and decrements the value of a shared counter variable using two threads. One thread (called *CounterIncrementer*) is responsible for incrementing the variable and the other (called *CounterDecrementer*) is responsible for decrementing the variable. *CounterIncrementer* increments the counter by 5000 while *CounterDecrementer* decrements the counter by 3000. Implement the threads using both the methods (extending thread class and implementing runnable interface) shown in class.

1. Print the final count value in the main thread without using join() method. Also, keep print statements in all the threads to demonstrate context-switching between the threads.
2. Use join methods to halt the execution of the main threads until execution of both the threads (*CounterIncrementer* and *CounterDecrementer*) are over and then print the correct value of counter.
3. Apply syncrhronization to print the final correct value.